

1. RESEARCH QUESTION

Does stock screening methodology affect the out-of-sample performance of cointegration-based statistical arbitrage?

- Cointegration-based stat-arb requires searching across many candidate asset groups.
- Exhaustive testing is computationally expensive.
- Screening can reduce the search space but may also change the characteristics of the candidates entering the cointegration stage.
- This study compares Full, PCA, and FF3 screening and examines their impact on out-of-sample performance.

2. RESEARCH DESIGN

Experimental Framework

- Universe: TOPIX 50
- Training: Jan 2025 – Dec 2025
- Test: Jan 2026 – Jul 2026
- Candidate groups: 2–3 assets
- Selected spreads: Top 5
- Only the screening methodology was varied; other major parameters were fixed.

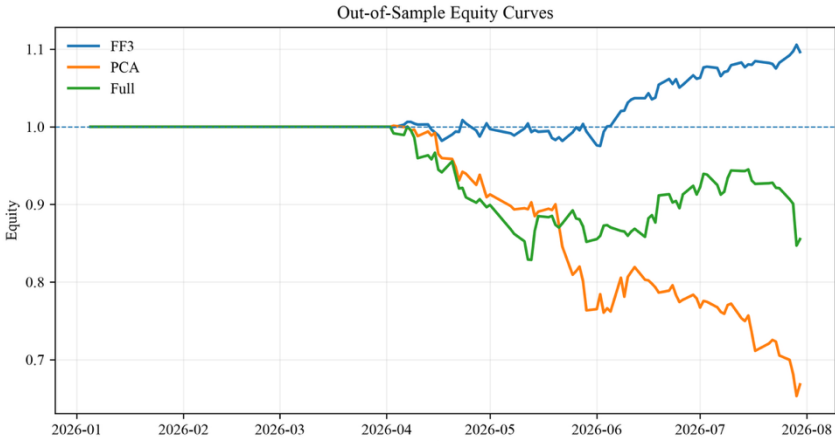
Screening Methods

Method	Representation
Full	Baseline; direct candidate search
PCA	Low-dimensional representation of training-period return patterns
FF3	α + Market + SMB + HML exposures from Fama-French 3-factor regressions

Cointegration & Trading

- Candidate groups → cointegration testing → 60-day persistence evaluation → Top 5 selection
- Spread model: Kalman filter
- Z-score window: 60 days
- Entry / Exit: 0.7518 / 0.25
- Entry threshold 0.7518 = analytical optimum under the Gaussian threshold model.

3. EMPIRICAL RESULTS



Selected Cointegration Spreads

Screening Method	Tickers
Full	5803-6301-7203, 5803-7203-7974, 5803-6301-9983, 6701-8035-9434, 6301-6762-9984
PCA	6762-7203-7741, 6762-6954-7741, 4502-4568-4578, 6752-7267-8031, 6752-7267-8591
FF3	5803-6758-6954, 6702-6861-7974, 4062-6501-6723, 4062-6501-6758, 6861-7974

4. INTERPRETATION

Key Interpretation

Screening methodology may affect the quality and robustness of the cointegration candidates ultimately selected for trading.

- FF3 outperformed Full and PCA in this experiment.
- Possible explanation: FF3 groups stocks by economically interpretable factor exposures, providing information distinct from direct return-pattern similarity.
- This may help identify cointegration candidates with stronger out-of-sample characteristics.
- However, this does not establish that FF3 is generally superior.
- The observed difference may also reflect the sample period, market environment, or other design choices.

Key takeaway

The design of the candidate-generation stage can propagate through cointegration testing and ultimately affect trading performance.

5. LIMITATIONS & FUTURE WORK

Limitations

- Single train/test split: limited evidence on regime robustness.
- Three screening methods: broader methodology comparison remains open.
- Fixed parameters: interactions with cluster size, z-score window, thresholds, etc. were not tested.
- No statistical significance / confidence-interval analysis.

Future Work

Direction	Representation
Robustness	Transaction costs, parameter sensitivity, multiple periods
Strategy Design	Improve cointegration candidate ranking
Computation	Accelerate Johansen testing and enable larger searches
Generalization	Other screening methods and markets, e.g. US equities